

An Analysis of Indian Stock Market Cycles and Demographics of Indian Investors

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Abstract

The purpose of the present study is to show the impact of demographic and behavioural factors on Indian individual investor's decision making in financial markets. The papers attempts to provide empirical evidences that the investment decisions take in the Indian household are largely driven by behavioural factors rather than financial postulations. Further, with the use of primary data, the study attempts to unearth the impact of bull and bear market sentiments on Indian investor along with identify seven bear phases and six bull phases in Indian stock market during the time period April 1995 till March 2016.

It has generally being said that when a big market breaks badly on good news it is a bear market; when a big market rises sharply on bad news it is a bull market³. You always find lots of early Bulls in Bull markets and early Bears in Bear markets.⁴ Both Bull and Bear markets trends always have a large influence over the investment decisions of investors. To understand how the external and internal factors of investors interact and which factors influence investor's investment decisions and strategies in Indian market primary survey is a valuable method. In past researchers have conducted surveys to understand investment decisions, behavioural determinants of perceived investment performance, effect of behavioural biases, factors affecting investment decisions, effect of demographics on investment decisions (Shabgou & Mousavi, 2016), (Khan, 2014), (Farooq & Sajid, 2015), (Kengatharan & Kengatharan, 2014), (Gupta, 1991), (Chandra & Kumar, 2011), (Chandra & Kumar, 2011) (Teeuwen & Huisman). In the present study, an attempt has been made to understand the behavioural factors affecting India investors and focuses on the psychological biases which influence individual investors' behaviour in Indian context. The study also creates dynamics of investment by creating and testing portfolios that can be formulated on the basis of behaviour of Indian investors.

Taking into consideration the theory of irrationality, researchers have shown that investors across financial markets do not act in purely rational manner, rather their investment decisions are influenced by a number of factors which also include psychological biases, heuristics, social affiliation, demographic factors and so on (Kumar and Lee, 2006; Baker and

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³ <http://www.nowandfutures.com/quotes.html>

⁴ Jesse Livermore, "How to Trade in Stocks" <http://www.nowandfutures.com/quotes.html>

Wurgler, 2007; Gärling et al, 2009; Barnea et al, 2010). The purpose of the present study is to show the impact of demographic and behavioural factors on Indian individual investor's decision making in financial markets.

Review of Literature

Obamuyi (2013) determine the main factors influencing investment decisions of investors and how these factors are related to the investors' socio-economic characteristics in the Nigerian Capital Market. The study found five factors having most impact on investment decisions of investors in Nigeria are past performance of the company's stock, expected stock split/capital increases/bonus, dividend policy, expected corporate earnings and get-rich-quick. Kengatharan & Kengatharan (2014) explores the behavioral factors influencing individual investors' decisions at the Colombo Stock Exchange along with identifying the relationship between these factors and investment performance. They found four behavioral factors which affect the investment decisions of individual investors which were Herding, Heuristics, Prospect and Market. The variables from all factors were mostly found to have moderate impacts while anchoring variable from heuristic factor has high influence and herding factor was found to have low influence on investment decision. (Shabgou & Mousavi, 2016) found Heuristic factors, prospect factors, market factors, herding effect factors have impact on investors' decision-making and that the market factors have the highest correlation with investors' decision-making. (Khan, 2014) explored the behavioral determinants of investment performance in Karachi Stock Exchange with risk perception as mediating variable while financial literacy has moderating variable. Their study found that two variable framing and herding had a positive and significant relationship with investment performance. However, mediation was not established and financial literacy showed moderating relationship with framing effect and perceived investment performance.

An exploration of the literature identifies that the behaviour and sentiments of individual investor acts as an important cause of movements in stock prices and subsequent returns. An identification of behavioural factors may help in re-examining the asset pricing theories by incorporating behavioural factors into the existing theories and portfolios of investors. Researchers in both developed and emerging economies have continuously tried to address the behavioural issues in stock market in different ways. Those who support behavioural finance try to explain using behavioural and psychological factors while those who believe in efficient market hypothesis, known as modern portfolio theorists, advances capital asset

pricing model or asset theories for formulating investing strategies. Investor behaviour is widely explored in literature using stock market data, however assessing the investor behaviour using questionnaire based studies supplements the investor dynamics. Behavioural finance researchers have adopted this approach to identify the significance of several cognitive and behavioural factors on individual behaviour. Past researchers have conducted surveys to understand investment decisions, behavioural determinants of perceived investment performance, effect of behavioural biases, factors affecting investment decisions, effect of demographics on investment decisions (Shabgou & Mousavi, 2016), (Khan, 2014), (Farooq & Sajid, 2015), (Kengatharan & Kengatharan, 2014), (Gupta, 1991), (Chandra & Kumar, 2011), (Chandra & Kumar, 2011) (Teeuwen & Huisman).

Kim and Nofsinger (2007) study individual investors in the Japanese markets and examine their behaviour and performance. Using the market level data they find that Japanese investors own risky, high book-to-market stocks, trade frequently, buy recent winners but make poor trading decisions. They also find that these factors appear to vary depending on the bull or bear market conditions. They observe that it is primarily during a bull market where individuals tend to hold high book-to-market stocks, as opposed to a bear market where they exhibit an inclination towards high beta stocks. Overall the poor performance by individual investors can largely be explained by this tendency to hold value stocks during advancing markets and high risk stocks during declining market.

Behavioral Aspects of Investing

Behavioral Finance was developed in late nineties as a new field of finance which gained popularity when Amos Tversky and Daniel Kahneman won the 2002 Nobel Memorial Prize in Economic Sciences, with their development of Prospect Theory. The concept of psychology plays a big part in investing and understanding the psychological motivations can help investors avoid financial pitfalls. Behavioral biases are the basis for understanding cultural differences, which in turn are the basis for understanding neurofinance and thus are fundamental to selecting a wealth management approach by investors. (Hens & Meier).

Behavioral finance a modern branch of finance helps to bridge the gap between financial theory and investment practice by methodically recording human behavior. Till late nineties, financial research has focused on rational investors in efficient markets, while reality deals with day-to-day irrational investor behaviors and inefficient markets. Combining theory and

practice allows us to use behavioral finance as the basis for advisory services, asset management, and financial product development.

Overconfidence

“In this most basic form, overconfidence can be summarized as unwarranted faith in one’s intuitive reasoning, judgments, and cognitive abilities” (Pompian, 2006). Psychologists have determined that overconfidence causes people to overestimate their knowledge, underestimate risks, and exaggerate their ability to control events. The concept of overconfidence derives from a large body of cognitive psychological experiments and surveys in which subjects overestimate both their own predictive abilities and the precision of the information they have been given. In short, people think they are smarter and have better information than they actually do (Pompian, 2006). According to Shefrin (2000), Overconfidence “*pertains to how well people understand their own abilities and the limits of their knowledge*” Individuals who are overconfident about their abilities tends to think they are better than they actually are.

Herding Behaviour

Herding in financial markets can be defined as mutual imitation leading to a convergence of action (Hirshleifer and Teoh, 2003). Herd behavior is the tendency individuals have to mimic the actions of a large group irrespective of whether or not they would make the decision individually. This is the most common mistake where investors tend to follow the investment decisions taken by the majority. That is why, in financial markets, when the best time to buy or sell is at hand, even the person who thinks he should take action experiences a strong psychological pressure refraining him to do so. The main reason for this is pressure from or influence by peers.

Disposition Effect

The disposition effect is defined as the tendency of investors to hold on to losing stocks and sell winning stocks early. This concept is built on the implications of prospect theory. The possible reasons for this effect as proposed by the authors are loss aversion, seeking pride, mental accounting and regret avoidance. They stipulate that realization of gains in individual stocks can make the investors proud. The major research work in the disposition effect took place in the late 1990’s. Researchers differentiate between overconfidence bias and the disposition effect where the former bias should be studied at market level and the latter bias

at the individual stock level. Their results are consistent with the disposition effect trading and the trading of overconfident investors in relation with trading volume.

Research Objectives and Hypothesis

In view of studies reported here above and evidences therein, the present study attempts to study the primary and secondary data to unearth the behavioural disposition of Indian investors, especially in bull and bear market runs. The study aims:

- To gather deep understandings about the two most prominent phases of Bull and Bear of the Indian stock markets.
- It intends to test whether individuals' attitudes and preferences toward stock risk, valuation, and past returns are different between market conditions.
- To examine whether internal (gender, age etc.) and external factors (Bull-Bear market conditions) interact to influence the disposition effect.
- To examine various investment strategy in Indian bull and bear market.
- To investigate the effect of bull and bear markets on different investment strategy of the market timers and passive investors

The above objectives are proposed to be achieved through following hypothesis:

- H₀₁:** There lies no difference between the preference of retails investors towards investing in Bull and Bear market on the basis of their demographics.
- H₀₂:** The behavior of individuals across bull and bear market conditions is asymmetric.
- H₀₃:** The risk taken and number of times investors earn profit is independent.
- H₀₄:** The alphas and/or betas in bull markets are not statistically different from alphas and/or betas in bear markets
- H₀₅:** The size based investment strategy does not yields extra normal returns in Bull and Bear phases
- H₀₆:** The value based investment strategy yields symmetric extra normal returns in Bull and Bear phase.
- H₀₇:** There is no extra return earned by formulating portfolios using Sharpe Index Single Model.
- H₀₈:** There is no size premium and value premium during bear or bull phases.

Data & Methodology

The present study was conducted using the survey method using a structured questionnaire which was developed after reviewing the literature. The questionnaire was floated through

online and offline mode was collected from 618 respondents around Delhi-NCR region. After cleaning the data 400 responses were used for empirical analyses. The target respondents were resident of Delhi-NCR region and that he/she should be 18 years or above and must invest or trade in Indian stock market with their own financial savings or invest/trade on behave of others. The respondents are selected from the Delhi- NCR region only because it is amongst the top metro cities actively trading in Indian stock markets. The study adopted convenient sampling technique to collect the survey responses and the sample was randomly drawn for the purpose of data collection.

Different statistical tests as required by the type of data and analysis were used such as Chi-square test, t-test analysis, one- way ANOVA. The reliability of survey instruments was verified with the help of Cronbach's alpha method which is a measure of internal consistency i.e., how closely related a set of items are as a group. The Cronbach alpha was found to be more than 0.6 for all items which is considered a good indication of construct reliability.

Factor Analysis was developed for the purpose of analyzing the inter-correlations within a set of variables. It may be used when we wish to reduce the dimensionality of a set of variables by taking the advantage of inter - correlations among variables. It helps to concentrate only on significant constructs & to avoid insignificant variables.

Survey Instrument: Questionnaire for Investors

Assessing individual behavior thorough questionnaire survey is a well adopted approach in behavioral sciences research (Chandra & Kumar, 2011). Survey research was undertaken to investigate behavioural factors and characteristic of investors with the help of a structured questionnaire.

This questionnaire is divided into 3 sections. Section I is based on demographics of the respondents such as age group, gender, marital status, highest educational qualification, professional experience, occupation has been sought in this section. This information will help to draw conclusion about investing pattern on the basis of demographic profile of respondents. Section II focuses on general investment behavior in stock market such as the investor's preference for time to enter/invest in the markets, formulating goals related to portfolio allocation, portfolio creation, percentage of annual income invested in stocks, the risk they like to undertake along with return they usually earn on their portfolio. Section III contains statements on investor behaviour biases on a five point Likert namely, 1 (strongly disagree) to 5 (strongly agree). Statements related to behavioural factors and attitude towards

investing/trading in rising/falling markets in Indian stock markets along with statements on Cognitive bias factors such as Disposition bias, Overconfidence, Trading Behaviour Self-attribute bias, Trading Attitude, Risk Tolerance, and Herd Behaviour. These factors were also confirmed through confirmatory factor analysis. The individual investor/trader behaviour is influenced by multiple factors but it is beyond the scope and affordability of this study with time limit to consider all those behavioral factors.

The content validity of the questionnaire was verified by expert guidance and discussion with three experts, two academicians and three industry professional has been sought. Accordingly the survey instrument was edited, eliminating few statements, rewording statements, replacing few technical words with simple words in the questionnaire. Thereafter a pilot study on 50 respondents has been conducted and tested whether results were achieved, before floating the questionnaire to target respondents.

Demographic Profile of Indian Investors

The respondent's demographic profile and investor sophistication is analysed through variables such as age group, gender, marital status, educational qualifications, professional work status and total professional experience. It also includes annual family income and the percentage amount invested in stock market. While the investor sophistication variable includes whether they trade or invest, their trading experience, their trading frequency, the market phase preferred, benchmark preferred, risk and return profile, corporation preferred while investing, *et al.* The percentage of respondents under age category showed that majority belong to the age group of 26-35 years followed by age group of 36-50 years, 20.38% are in the age group of 18-25 years and only few were above the age of 51 years. Also, 24% of respondents were females, out of which 69.8% of them were married while 76% were males with 67.4% of them being married.

Amongst the respondent's current work status, 35.3% of them were working as service class, followed by 24.5% working as stock broker/analyst, 18.3% were CA/CS/Lawyer, mere 2.5% were academicians, 8.5% were businessman, 5.8% were students and 5.3% were retired individuals. Majority of females respondents are presently working as Service followed by stock brokers/analyst. Hence, this suggests that the females are less interested in investing in stock market and those who are finance graduate don't stay away from the stock market but work as stock brokers also and as they are financially independent invest their own money in stock markets for long-term gains and have stock market experience of more than 5 years.

Thus females as generally thought but do not stay away from the risky stock market but are attracted towards the market to majorly invest their hard earned money in stock markets.

Also, the majority of respondents had more than 10 years of professional experience followed by 22% professional experience of 7-10 years. Also, majority of the data pertains to the samples who are working but only one fourth of the sample works as stock brokers/ analyst. Hence, we may state that it is not necessary that only those who work directly in stock market tend to use it as investment avenue and education has helped to change this conventional attitude.

The respondents mostly belong to middle income group followed by high net worth respondents. It is seen from the survey that those who belong to low level income trade in stock market more than those in middle or high level income group who invest their money in stock markets to gain higher returns. Also, those with low income just enter the market for short term period when the markets are rising and sentiments are positive but they don't invest for long term in the stock markets. This gives further push the bubble in bull markets, therefore incentives must be created for the retail investor to stay in the market for long term and provide stability to the market. This is confirmed with the responses data which shows that majority of those belonging to middle and high level income classifies themselves as those who use their own money to purchase equity shares. Also, irrespective of income group they belong to majority prefer to invest in rising markets or bull market while only one-fourth of high income group also invested in falling markets.

Also it was found that not many investors with low income prefer to put their hard money in stock markets. This is in confirmation with low proportion of retail investor's participation in Indian stock market.

Indian Investor's Trading Sophistication

The investors trading sophistication helps to gauge the general behaviour of Indian investors which decides their trading behaviour and investment pattern. The results show that more than three-fourth of Indian investors prefer to invest in rising markets, i.e. Bull markets when the stock prices are rising and economy is strong with hope of continuing optimism in the economy. While merely one-fourth of the respondents prefer to invest in falling markets, i.e. Bear markets when the prices are falling on the hope to earn higher profits when prices increase in future. Also, the majority of respondents are individual retail investors while only one-third respondents are traders.

It's seen from the survey that less than one-fifth of females trade in Indian stock market compared to more than four-fifth of total males traders. Chi-square test statistics shows that there is statistically significant difference between gender and their preference to trade or invest. Males and females statistically differ on their preference to invest or trade with chi-square value of 46.240 significant at 5% level with p -value of $.000 < .05$.

The survey show that majority of respondents who are in their late-30's have high trading experience of more than 5 years while those who are in their mid-20s to mid-30s have 3 to 5 years of experience. This shows that with increase in internal factor, i.e. age of an individual the trading and investment in stock market increases as the motive to earn higher return increases with age as more responsibilities lies on the individual. Thus with increase in age, trading experience increases and they learn from their past mistakes due to which after 36 years of age the activity in stock market increases. Hence, these investors should be captured to stay invested in the market for longer duration which will give boost to the market and stability to the index level. The survey results also show that Indian investors preferably indulge in stock market activities monthly or yearly basis to evaluate their past year investments and returns as they enter the market for long term investments and don't change their holding very frequently.

With more trading experience and more frequent market activity the expected returns increases and also lead to behavioural bias such as self-attribution bias⁵. The chi-square analysis of associations shows that there is a statistically significant relation between trading experience and frequency of stock market activity. Pearson Chi-square value of 72.946 is statistically significant at 5% significant level with p -value of $.000 < .05$. Thus respondents with higher trading experience trade more often (on daily basis) than those who have less trading experience and therefore trade less frequently in stock market.

Using the One-Sample Kolmogorov-Smirnov Test of Goodness of Fit test it seen there is statistically significant difference at 5% that Indians prefer Sensex more than Nifty as the benchmark index in Indian stock market. This is despite the fact that Nifty has been far more transparent since its beginning and has stronger future growth compared to Sensex.

Risk and Return Profile of Indian investors

⁵ The self-attribution bias states that investors attribute successes to their own skills and attribute failures to bad luck (Teeuwen & Huisman, n.d.).

The respondents risk undertaking ability and times they earn profit from their investment was ranked from 1 to 10, whereby 1 is low and 10 means high. It was found (table 1) that most investors prefer to take medium risk with mostly earning medium return from their investment whereas less than one-third investors take high risk but earn high return only 15% of time and only just 7% take very high risk but earn profit only 3.3% of times from investment.

Table 1 Risk-Return Profiles

Rank		Risk	Return
Low	1-3	22.8%	36.3%
Medium	3-5	41.0%	45.0%
High	5-7	29.0%	15.5%
Very High	7-10	7.2%	3.3%

Source: Research Findings

The direction of correlation was found to be positive implying that those who undertake high risk tend to earn high profits through stock market activities. The value of Phi, which tells strength of association between variables, is high (or 51.7%) and strong between risk and return rank. Thus, the ‘high risk, high return’ mantra is confirmed in our analysis.

Phase: The analysis of interaction between phase preferred (bull or bear) with the risk and return ranks shows that irrespective of which phase is preferred, Indians undertake medium risk with mostly earning medium return on an average as shown in table 2. However, the chi-square result were not found to have statistically significant relation between the bull and bear market phases preferred and risk taken scores and return earned score

Table 2 Bull and Bear Markets and Risk and return ranks

Risk Ranks	1-3	3-5	5-7	7-10
	Low	Medium	High	Very High
Bear Markets	20.6%	43.3%	26.8%	9.3%
Bull Markets	23.4%	40.3%	29.7%	6.6%
Return Ranks	1-3	3-5	5-7	7-10
Bear Markets	30.9%	49.5%	17.5%	2.1%
Bull Markets	38.0%	43.6%	14.9%	3.6%

Source: Research Findings

This shows that the preference for bull or bear phases doesn’t have influence on the risk taking and return earning behaviour of Indian investors.

Gender: The Mann-Whitney U test found statistically significant difference in the risk and return scores on the basis of gender. The result (table 3) shows that male mean risk score and return earned rank is higher that of females which implies that males are more risk-lover and high risk taker thus earning high return. This also confirms that males have more confidence than females when it comes to taking risk.

Table 3 Risk and Return Vs Gender

Ranks	Gender	Mean Rank
Risk_Undertaken	Female	133.89
	Male	221.53
Times_Profit_Earned	Female	176.11
	Male	208.20
TEST STATISTICS		
Test Statistics ^a	Risk_Undertaken	Times_Profit_Earned
Mann-Whitney U	8197.500	12250.500
Asymp. Sig. (2-tailed)	.000**	.010**
a. Grouping Variable: Gender ** Significant at 5% level.		

Source: Research Findings

Age: The results of Kruskal Wallis test found statistically significant difference between different age group of respondents and risk taking behaviour and return earning score. The result shows that Mean rank for age group 36-50 is highest for both risk undertaken and profit earned in stock market which means that those who are mature take higher risk and thus earn higher return than their younger and older generations.

Professional experience and Stock Market experience

The result of Kendall's tau-b found a significant positive association between professional experience and experience in investing or trading in stock market, tau (400) = .641, *p*-value < .05. Thus with higher professional experience, the interest to invest or trade in stock markets increases.

Also a strong statistically significant correlation was found between risk score ranks, profit earning ranks, professional experience and experience in stock markets activities which implies that those with higher professional experience usually have higher stock market experience and they tend to take high risk to earn high return from stock market activities.

Table4: Correlation between Risk, Profit earned, Professional Experience (in years) and Stock Market Experience (in years)

Correlations ^b				
	Risk Ranking	Ranking of Profit earned	Professional Experience (in years)	Stock Market Experience (in years)
Risk Ranking	1			
Ranking of Profit earned	.433** (.000)	1		
Professional experience (in years)	.278** (.000)	.162** (.001)	1	
Since how many years you have been trading or investing in Stock Market:	.333** (.000)	.204** (.000)	.780** (.000)	1

** . Correlation is significant at the 0.01 level (2-tailed).

Source: Research Findings

Behaviour in Bull and Bear Phases

Investors and traders have different preference for bull and Bear phases have been found in previous studies mainly in US, Japan, Europe markets (see for eg. Kim & Zumwalt (1979), Barsky & Long (1990), Chen (1982), Chi, Brooks, Bissoondoyal-Bheenick, & Tang (2012), Kim & Nofsinger (2007), Gonzalez, Powell, Shi, & Wilson (2005), Maheu & McCurdy (2000), Pagan & Sossounov (2003), Tu (2005) *et al*). These studies have been conducted using secondary data to analyse the preferred phase and behaviour of individual in these phases and also these studies are undertaken in USA, Europe, Japanese, Chinese stock market. However, not many studies have been conducted using primary data and nor much in Indian context.

This section attempts to gauge the behaviour and risk taking pattern and return earning behaviour in Indian bull and bear markets. It is of importance to know whether behaviour in a rising stock price market, i.e. a bull market could also be exhibited in a declining price stock market, a bear market. The present study shows that the behaviour of individuals changes in different market conditions according to their demographic profile and external factors.

The study result show that majority (75%) of individuals prefer to invest in Bull market, i.e. when the stock prices are rising with continuous hope of price rises and growth in economy. Also, the results show that both the gender equally highly prefer Bull markets more than Bear market for investing and trading. Thus, the preference of Indian investors is more towards Bull market, during which the domestic retail, institutional and FIIs participation and investment also increases in Indian stock market. This is also established from the results that both the respondents who invest and trade highly prefer Bull market when the chances to make higher profit is more. In each Indian stock market cycle, individuals and traders of different age group highly prefer rising markets which also show that preference to earn high returns and to invest or trade in the stock market is more than in Bear market. The euphoria that is created by northward movement of stock prices keep everyone interested in the stock market no matter the age, gender or type of individual- trader or investor. As it can be inferred from these results that even those with higher stock market experience, still feel more comfortable dealing in rising markets when the chances of making loss is less as compared to falling markets. It is easier to time the market and get profitable results from the different investment strategies in Bull markets than in Bear market when everyone is fearful of the future changes. Thus with high experience and upward learning graph of high experience in stock market the preference for Bull market doesn't reduces. Similar, results

can be seen with high professional experience in their career, individual prefer rising markets more than falling market to protect themselves from losing their money in market crashes.

The results of preferences to invest in the two market shows that merely one-fourth of the respondents prefer to invest/trade in falling markets, i.e. Bear markets, when the prices are falling but they trade/invest on the hope to earn higher profits when prices increase in future. This is in confirmation with previous studies that most individuals and organization put their money in stock market which is thought to be otherwise a risky investment, only the market sentiments are positive, prices are rising and economy is growing.

The frequency of stock market activities of an individual also helps to understand which market state the investor or trader would be willing to put their money in. The study shows that about fifty percent of those who prefer Bear markets invest/trade on yearly basis in the stock market. Similarly those who prefer rising market, i.e. Bull market deal in stock market activities frequently on monthly basis followed by those who deal on daily basis. This implies that long term investors having investment horizon of a year, buy stocks in Bear period when stock prices are falling to invest money for long-term gains. While those who trade for short horizon such daily basis or weekly are more interested in rising markets. There is a significant difference in preference for Bull and Bear markets on the basis of respondents trading frequency.

It can be seen that Bull and Bear markets due to their different characteristics and dynamics are preferred differently by different type of stock market participants. Those who individuals who use their own money to purchase equity securities and those engaged in the trading of equity securities both highly prefer Bull markets and are more willing to put the money in rising markets. This can have important implications for corporates, financial institutions and governments to take actions to bring more retail investors to the market during Bull markets and make efforts so that those who come to the market at this time doesn't come in just to leave when the sentiments are bad. As more people in the market during Bear markets can have stability effects on the prices and index level which can strong our stock markets in the world economies. But it is often seen that while in time of crisis in the world or domestic crisis, investors leave the market after losing their money during bear market. This leads to further exaggeration of the negative sentiments leading to fall in stock prices and index level. Similar mutual fund investors and systematic investment plan investors must also be given incentives to enter the market during Bear markets when prices are low but sentiments are negative as this will give a push to the market, thus increasing the investment flow to the

market and increasing the stock demand. This would help the economy to grow and corporates to invest more in projects that will give a boost to the whole economy. It is seen from the study that those who prefer falling markets, around 57% of those are individual who use their own money to purchase equity securities thus, the fund managers should come out with profitable funds and strategies to attract and retain this class of investors to change the market sentiments from Bear to Bull by giving an upward push to the market index level.

Risk Profile

Investing in stock market is perceived to be a risky investment by many which keeps them away from the stock market. Risk taking behaviour determines the return of a person as there is always a trade-off between risk and return. Higher the risk an individual takes, higher the probability of earning high return and higher the probability of incurring high losses. The respondents were asked to rank on a scale of 1-10 their risk undertaking ability and also similarly rank the times they earn profit from their investment (whereby 1 means low and 10 means high). The results of the study show that Indian investors are mostly medium risk takers followed by high risk takers as seen in Indian respondents. On the basis of risk taking behaviour, study found no significant relation between risk taking and preferred market phase. Also irrespective of risk taking behaviour all respondents prefer to invest in Bull market. However, the proportion of those who prefer falling markets but take medium risk is higher than those who prefer rising markets. Similarly those who take high risk and prefer falling markets is more than those who take high risk but prefer rising markets. This shows that individuals and traders who are risk-lovers prefer to invest/trade in Bear markets to earn short-term profit during Bear rally and to invest for long-term by buying stocks at lower prices during Bear markets. Thus the fund managers should try to tap the high risk lover investors during the falling markets.

Individuals compare their investment performance with the return earned by benchmark index such as Sensex. It is seen in the present study that those who prefer Bear market assess their own investment performance more frequently as over-performing as compared to that of Sensex. While those who prefer Bull market assess their investment performance as under-performing compared to Sensex. This implied that people tend to be over-confident about their investment in Bull market and may make mistakes which lead them to earn lower return compared to Sensex returns.

Similar results are seen in terms of the corporation preferred by investors and traders in both the Bull and Bear markets. Individual investors in highly uncertain markets prefer to take less risk and invest their money in corporation which in past have given high and stable returns.

Thus, large corporation are most preferred investment corporations in both rising markets (Bull markets) and falling markets (Bear markets).

In past studies such as Pagan & Sossounov (2003), Gonzalez, Powell, Shi, & Wilson (2005), Statman, Thorley, & Vorkink (2006), Glaser & Weber (2007), Sornette (2003) *et al* it was empirically tested whether trading volume have any effect on investment decisions of investors. Prosad (2014) in their doctoral result synopsis have studied that trading volume have effect on behavioural bias such as Disposition effect, over-confidence in investors. The respondents of the present study also confirms to the fact that the changes in trading volume of stocks and index levels does affect their investment decisions in both Bull and Bear markets. Thus, our study adds to the present literature that behaviour of individual in Indian stock markets is related to level of volume transaction at a particular time.

The results show that Indian investors in both the Bull and Bear markets prefer to invest using Technical analysis more compared to fundamental analysis. Fundamental analysis is more often used by value investors who give their time and energy to evaluate the intrinsic value of the firm before investing their money in those stocks. But as this involves active involvement to make investment decisions, many investors leave this task to the fund managers and follow technical indicators such as past volume, historical prices and formation, before making their decisions to invest. On analysing the different types of fundamental ratio preferred by individuals, it was found that Price-earning (P/E) ratio is the most preferred followed by Book-value to Market-value ratio which is often taken as a benchmark to formulate investment strategies. Historical P/E ratio and BV/MV ratio gives an indication about future performance of the stocks which helps the investors in Bear markets to make profitable strategies which will help them earn better results. In Bull markets, P/E ratio are used followed by Market Capitalisation of the company is preferred by individuals while making investment decisions. Debt-to-equity ratio and Dividend-Payout ratio of the stock is not much preferred by Indian investors while making decisions. This can be due to the reason that investors look to earn return from capital gains that is rise in stock prices rather than dividend incomes as there is many-fold increase in stock prices compared to their intrinsic value. Thus, market based fundamental ratio is more preferred by individuals which should be the focus of the corporates while making an corporate investment decisions which can have effect on the market value and market ratios of their firms in both Bull and Bear markets.

Stock markets gain upward movements followed by downward movements leading to crashes. Individual's behaviour and reaction to market positive high jumps and crashes helps

to understand their behaviour in different phases of the stock market cycles. The present study found similar behaviour of individual investors and traders in market crashes such as their reaction to fall in Sensex in 2008 due to financial crisis (US sub-prime crisis) that negatively affected all stock exchanges around the world. The result show that equal proportion of individuals purchased more shares at low prices or sold off shares during this period. Those with medium to high risk purchased more shares at low prices while those who take very low risk to medium risk sold the shares and exited the market during such big market crash and held more in safer investments like Government bonds, PPF, etc. Thus, investors are cautious of putting their money in stock markets, when the world markets are crashing down due to macro-economic variables and failure of sophisticated financial products, leading to market crashes, which are further intensified due to retail investors pulling out their money due to fear of high losses from the markets. During such crisis period, they prefer to hold their money in cash or in safer investment. Those who are young and more risk lover, also take this opportunity to enter the market and purchase shares at lower prices to take benefit when the market rises after the crash. This is because the market cycle alternate from period of crashes to period of high positive sentiments which again takes the index level to cross previous high level.

Disposition effect is when an individual investor sells a stock in his portfolio, he has a greater propensity to sell a stock that has gone up in value since purchase than one that has gone down. Barberis & Xiong (2009) explains how Prospect theory explains the model of Disposition effect wherein if an investor is holding a stock that has risen in value since purchase, he may think of the stock as trading at a gain. If he is risk-averse over gains, he may then be inclined to sell the stock. Similarly, if he is risk-seeking over losses, he may be inclined to hold on to a stock that has gone down in value. Thus, to analyse if Disposition effect is present among Indian investors, the respondents were asked their reaction on an event of a significant loss due to downturns in the market, how long they are prepared to hold existing investments in anticipation of a recovery in value. It is found that Indian investors prepared to hold on to their shares for six months to one year in event of significant losses in anticipation of rise in stock prices within next one year. Also, most are prepared to hold on to their shares in even of possibility of significant losses for just three to six months after which they sell their shares and hold more in cash or safer investment when the possibility of huge fall in stock prices increases with duration.

Factors Affecting Investment Decisions

The study also found following top five factors which affect investment decisions of investors:

1. Capital Appreciation as per your expectations, **mean score 4.4**
2. Macro-economic variables like Interest rate, growth rate of GDP, oil prices, exchange rate, inflation etc. **mean score 4.18**
3. Past yearly performance of the stock, **mean score 4.03**
4. Current market phase, whether it is increasing or decreasing, **mean score 3.85**
5. Future expansion plans of companies **mean score 3.56**

The top five factors show that capital gain is the main factor that influences the investment decisions of Indian investors. Investors and traders prefer appreciation of their investment as main factor which influence their investment decisions. This implies that investor prefer return and therefore prefer to invest in rising markets when the probability of capital appreciation is high. The second most factor that affects investment decisions are macro-economic variables like interest rate, growth rate of GDP, exchange rate, inflation etc. that may influence their decisions. It shows that the stock market level is affected mostly by macro-economic variable which also in turn affects retail investors and traders decisions. The third factor that affects investors is past yearly performance of stocks which implies that the investors base their decisions on historical price and trends in stock prices to make future decisions. Future is always uncertain however, historical prices help the investors to calculate future expected return on their investment. The fourth important factor that affects investors in stock market is the underlying current phase of the market which is either decreasing or increasing. Thus, depending on whether the current trend is upward or downward investors may buy or sell the stocks or hold more in cash or in safer investments like gold, government bonds, etc. Thus, the current underlying phase is an important consideration in Indian stock market.

The fifth factor that affects investor's decisions is *Future expansion plans of companies*. The future expansion plans of the company helps the investor to analyse whether the return on investment in the stocks of the company will be more than the stocks of other companies. If the company announces its plans to expand in future then the company will retain the profit and not distribute as dividends, leaving less current income (dividend income) to investors. Thus if a company is not paying dividends and have no expansion plans then the investor will be able to earn more return on other investment. If a company announces expansion plans

during bearish periods this would imply the company fundamentals are strong and more investors will be willing to invest in stocks of that company.

Behavioural factors Influencing Indian investors

Proponents of Behavioral finance have argued that investors don't behave rationally when it comes to risk and uncertainty as their behaviour deviate from the behaviour of rational investor. Investment decisions are often influenced by various factors which have been widely discussed. A well-known economist and researcher, John Maynard Keynes asserted that nearly a hundred years ago: "The markets can remain irrational longer than you can stay solvent". Thus the irrationality exists in the behaviour of investors which can be influenced by behavioural factors such as herd behaviour, overconfidence, etc. Behavioural finance is new concept in modern finance which was propagated after the development of Prospect Theory by (Kahneman & Tversky, 1979), winners of the 2002 Nobel Memorial Prize in Economic Sciences.

It can be seen that in reality market participants are irrational however the theory and research has always focused on rationality of investors as also in Efficient Market Hypothesis. The prospect theory focuses on the significance of investment loss and that investors have aversion to loss. This implies losses from investment must be compensated by opportunity to earn high returns, i.e. the returns from their investment should be at least twice as high as potential loss. Hence, the prospect theory which asserts that investors compare their investment in a reference point and are on average risk averse by choosing prospect theory to base their portfolio can stick to their strategy both financially and emotionally.⁶

Thus as seen above, recent literature has shown that various behavioural factors influence the investment decisions of investors. However, not many studies to study the behavioural bias and factors that influence the Indian investor's behaviour have been conducted in reference to Indian investors. The present study adds to the existing literature as it tries to identify the behavioural factors which may influence the investor's investment decision on the basis of their preference to invest in either bull or bear phases.

The individual investors who are believed to be irrational are influenced by psychological biases. This bias affects financial decisions and thus their trading behaviour. It thus important to identify the factors which most affect the investors trading behaviour based on their internal and external profile.

⁶ CREDIT SUISSE Private Banking North America- Finance White Paper. Research conducted by Professor Dr. Thorsten Hens and MSc. BA Anna Meier from Behavioral Finance Solutions GmbH "Behavioral Finance: The Psychology of Investing"

The questionnaire contains 35 statements about Perception about Stock Market, Attitude towards Investing (trading) in rising/falling markets in Indian stock markets, trading behaviour, Cognitive factors during Rising/Falling markets, Attitude Towards Stock Risk, Valuation and past Returns; disposition effect.

Results of Factor Analysis

The exploratory factor analysis (EFA) was used to detect the behavioural factors which most influence the investor's behaviour. For the EFA research the following criteria are applied to detect the relevant factors: Factor loadings, KMO test and Bartlett test of Sphericity, Total variance explained, and Eigenvalue.

The Exploratory Factor Analysis is conducted using IBM SPSS software version 21. The results of Kaiser-Meyer-Olkin Measure of Sampling Adequacy measure value is 0.941 which is a higher value which implies that the sample data is sufficient to conduct factor analysis. Also, the Bartlett's Test of Sphericity has a p-value of $.000 < .05$ which makes us to accept the alternate hypothesis and reject the null hypothesis that the variables are uncorrelated in the population. Thus the null hypothesis is rejected which makes factor analysis suitable for the survey results.

Cronbach's Alpha is calculated to test the internal consistency and reliability of measurements of continuous variable on 5-point Likert measurement. . Cronbach's alpha should be at least 0.7 as per Nunnally (1978, p.245) which tests the reliability of measurement which is included in the factors formed after factor analysis.

The statements from 1 to 35 of the questionnaire were intended to explore the levels of Behavioral variables' impacts on the individual investment decisions in Indian stock market. After various rounds of calculating Cronbach's alpha by removing unsuitable statement the further analysis was carried on. The results shows that on deleting five statements the Cronbach's Alpha increased after which further analysis were performed. The remaining statements were analysed on the basis of their factor loadings and total variation explained. Thus, after grouping the statements seven factors were found with Eigen value more than 1 and with total variance explained 56.954 % of variation in the variables with the factor loading more than 0.5. Table 5-7 shows the result of exploratory factor analysis.

The cumulative variance explained is 56.954% which is more than 50% of variance which the factors explain in the investment behaviour of Indian investors.

Applying EFA on the statements, seven factors were extracted and these with their factor loadings are given in table 6. The factors which were extracted after careful examination of statements are Trading Behaviour, Self-Attribution, Trading Attitude, Disposition Effect, Confidence, Risk Tolerance, Herd Behaviour.

Table 5 Factors and Total Variance explained for factors

No	No. Of Items	Factors extracted	Eigenvalues	Proportion of Variation explained	Cumulative Variance explained %
1	5	Trading Behaviour	11.746	11.206	11.206
2	5	Self-Attribution	2.017	10.015	21.221
3	6	Trading Attitude	1.640	9.370	30.591
4	3	Disposition Effect	1.365	9.322	39.913
5	2	Confidence	1.122	6.440	46.353
6	6	Risk Tolerance	1.041	6.054	52.407
7	3	Herd Behaviour	1.003	4.547	56.954

Source: Research Findings

Table 6 Factors extracted and their Factor Loadings

Factors	Variables	Factor Loadings
SELF ATTRIBUTION	* I state my investment objectives in terms of expected returns only not risk during rising markets.	.604
	* I like taking risk during falling markets if my investment horizon is long.	.671
	* I don't take risk if I am trading for short run.	.592
	* I look at long term while defining my expected return.	.552
	* I am more concerned about the current market trend if my investment horizon is short period say less than 1 year.	.605
TRADING ATTITUDE	* I invest in companies having strong fundamentals with long term stable growth during rising markets.	.685
	* I buy past winners and sell of past losers during falling markets.	.628
	* I sell past winners and buy losers during rising markets.	.492
	* I start selling as soon as markets show fluctuations due to fear of loss.	.614
	* I prefer to hold high book-to-market stock during falling markets.	.628
TRADING BEHAVIOUR	* I prefer to buy small size company's stock during rising markets	.525
	* I hold more in stocks if the Sensex returns are expected to be higher than return on government securities.	.605
	* I hold more cash in down markets.	.656
	* I have earned negative returns during rising markets.	.477
	* I have in past earned positive returns during declining markets.	.578
DISPOSITION EFFECT	* I prefer to invest my money in safer assets during market declines.	.565
	* I prefer to make profits or at least break even in all trades.	.640
	* I sell stocks when there is increase in prices rather than decrease in price.	.790
	* I am more sensitive to losses than to gains in trading.	.506
	* I have the skills and knowledge required to outperform the markets.	.786
CONFIDENCE	* I remain confident of my portfolio even if am losing some money during falling markets.	.645
	* If the stock price fall by 20% do nothing and wait for investment returns to improve.	.486
RISK TOLERANCE	* If the stock price fall by 20% buy further investments to take advantage of lower prices, in anticipation of future gains.	.635
	* If your investment rises by 10% you expect the investment to keep rising for less than 6 months.	.615
	* If your investment rises by 10% you expect the investment to keep rising for more than 6 months.	.526

	* If your investment falls by 10% you expect the investment to keep falling for more than a year.	.712
	* If the stock price falls by 20% you sell the stock to avoid further worry.	.529
HERD BEHAVIOUR	* I invest on basis of rumor irrespective of what I think about the stocks.	.473
	* I get influenced by what others are doing & saying.	.657
	* I behave in similar pattern whether the market is a rising or declining market.	.651
Extraction Method: Principal Component Analysis.		
Rotation Method: Varimax with Kaiser Normalization. a. Rotation converged in 10 iterations		

Source: Research Findings

Reliability Test using Cronbach's Alpha for each factor

The Cronbach's Alpha is calculated for each factor to test the reliability of items included in the factors identified in the factor analysis. The results of Cronbach's alpha test are shown in the Table 7. The reliability test is done to confirm that the measurements are reliable for further analysis. The Cronbach's Alpha for each factor derived ranged from .545 to 0.817 which is considered high to conduct further analysis.

Table 7 Reliability Test using Cronbach's Alpha

Factor No	Construct	Cronbach Alpha	No Of Items
1	Trading Behaviour	0.785	5
2	Self-Attribution	0.773	5
3	Trading Attitude	0.817	6
4	Disposition Effect	0.628	3
5	Over-Confidence	0.545	2
6	Risk Tolerance	0.816	6
7	Herd Behaviour	0.639	3

Source: Research Findings

Perception of Indian Investors in Bull and Bear Market States

The impact levels of Behavioral variables on the investment decisions are identified by calculating the values of sample mean of each variable. In similar, the variables of investment performance are scored by identifying the mean values as given in Table 8 below. As the statements were 5-point scales are used to measure the impact levels of these variables. The mean values of these variables are calculated which assesses the impact levels of factors on the investment decision making.

Table 8 Factors Extracted and their mean score

Factors	Statements	Mean Score
Cognitive factors during Rising/Falling markets		
SELF ATTRIBUTION	I state my investment objectives in terms of expected returns only not risk during rising markets.	3.62
	I like taking risk during falling markets if my investment horizon is long.	3.57
	I don't take risk if I am trading for short run.	3.42

	I look at long term while defining my expected return.	3.61
	I am more concerned about the current market trend if my investment horizon is short period say less than 1 year.	3.69
	Total	17.91
TRADING ATTITUDE	I invest in companies having strong fundamentals with long term stable growth during rising markets.	3.44
	I buy past winners and sell of past losers during falling markets.	3.54
	I sell past winners and buy losers during rising markets.	3.3
	I start selling as soon as markets show fluctuations due to fear of loss.	3.38
	I prefer to hold high book-to-market stock during falling markets.	3.36
	I prefer to buy small size company's stock during rising markets	3.67
	Total	10.55
Attitude towards Investing/trading in rising/falling markets in Indian stock markets		
TRADING BEHAVIOUR	I hold more in stocks if the Sensex returns are expected to be higher than return on government securities.	3.32
	I hold more cash in down markets.	3.23
	I have earned negative returns during rising markets.	3.5
	I have in past earned positive returns during declining markets.	3.35
	I prefer to invest my money in safer assets during market declines.	3.47
	Total	16.87
DISPOSITION EFFECT	I prefer to make profits or at least break even in all trades.	3.53
	I sell stocks when there is increase in prices rather than decrease in price.	3.27
	I am more sensitive to losses than to gains in trading.	3.75
	Total	10.55
OVER-CONFIDENCE	I have the skills and knowledge required to outperform the markets.	3.98
	I remain confident of my portfolio even if am losing some money during falling markets.	3.71
	Total	7.69
RISK TOLERANCE	If the stock price fall by 20% do nothing and wait for investment returns to improve.	3.53
	If the stock price fall by 20% buy further investments to take advantage of lower prices, in anticipation of future gains.	3.16
	If your investment rises by 10% you expect the investment to keep rising for less than 6 months.	3.24
	If your investment rises by 10% you expect the investment to keep rising for more than 6 months.	3.7
	If your investment falls by 10% you expect the investment to keep falling for more than a year.	2.92
	If the stock price fall by 20% you sell the stock to avoid further worry.	3.49
	Total	20.04
HERD BEHAVIOUR	I invest on basis of rumor irrespective of what I think about the stocks.	3.6
	I get influenced by what others are doing & saying.	3.42
	I behave in similar pattern whether the market is a rising or declining market.	3.17
	Total	10.19

Source: Research Findings

The total of mean score in table 8 show that out of seven factors extracted, Risk Tolerance has the highest total means core. This implies that the risk tolerance as a behavioural factor have the highest impact on the investment behaviour of Indian investors. The risk taking behavior highly impacts the investment pattern which is followed by self-attribution factor impact on investment decision. The third factor is disposition effect which implies Indian investor shows high disposition in their investment pattern whereby investors sell shares whose prices has increased while keeping assets that have dropped.

The factors influencing investment decisions of investors were checked if they differ significantly according to age, gender, marital status, market phase, trading frequency, professional experience, investor or trader and trading experience. Table 9 contains the result of ANOVA conducted to find if any relation exists between internal and external factors. The results show that whether the respondent is an investor or trader has no effect on the behavioural factors. One of the reason for this maybe that there is a thin difference between an investor and traders. As an investor can be a trader also with taking the same decisions and making the same mistake as that of a trader. While a trader can also think like an investor with making investment for long term and thinking like an investor while marking his responses? The results also show that the trading frequency of an investor (trader) doesn't have any impact on the effect of behavioural factors on investment decision making in stock market. Thus more experience which is an external factor of investor and keeps increasing with time doesn't always led the investor to make informed decisions or to make trading mistake.

The Gender of a person which is an internal factor and remains fixed overtime has a significant relationship with Trading Attitude factor. The trading attitude and behaviour of male's vs females is found to be different in most researches. The result of this study shows that gender has different perception relating to attitude towards trading/investing in stock market. The male prefer to take more risk on an average than females and the analysis in previous section show that females more of an investor than trader in Indian stock market. Females enter the market to make long term investment.

The Age as an internal factor has an effect only on the *Risk Tolerance* of an individual. It is seen in past researchers that age plays an important role in deciding how much risk an investor is willing to take. It is seen that young and mature individuals are more risk taker while older group of people take less risk and thus earn less return on their investment. Thus, with age the risk tolerance level of individual decreases.

The Marital status of an individual is an external factor which doesn't remain fixed over time. Those who are married generally belong to social groups and clubs and thus are often influenced by the behaviour of the group they belong to or want to be recognized with. Those who are unmarried also belong to social groups at office, college or clubs, thus are also influenced by the behaviour and investment decisions and factors affecting their decisions come from the group. Thus it also seen from the ANOVA test of marital status with all the factors identified and the results show that marital status of a person is associated with his/her

Herd Behaviour. Herd behaviour is the act of following what the herd is doing and thinking that whatever decisions they make is the best. Thus those who belongs to group often don't take the decisions they think to be right but go with the herd behaviour. No other internal or external factors were found to have significant relation with Herd Behaviour bias.

Professional experience also was found to have no relationship with internal or external factors of Indian investors and traders. Thus the more professional experience an individual gains make no difference to his/her investment decisions and factors that influence them.

Table 9 Result of ANOVA with factors extracted

No	No. Of Items	Factors	Mark et Phase	Trading Freq.	Invest or/ Trader	Age	Marital Status	Gender	Professional Experience	Trading Experience
1	5	TRADING BEHAVIOUR	-	-	-	-	-	-	-	-
2	5	SELF ATTRIBUTION	-	-	-	-	-	-	-	-
3	6	TRADING ATTITUDE	***	-	-	-	-	***	-	-
4	3	DISPOSITION EFFECT	***	-	-	-	-	-	-	-
5	2	OVER-CONFIDENCE	***	-	-	-	-	-	-	-
6	6	RISK TOLERANCE	***	-	-	***	-	-	-	-
7	3	HERD_BEHAVIOUR	-	-	-	-	***	-	-	-

Source: Research Findings

Investment strategies in Bull and Bear Phases

Both Bull and Bear markets trends always have a large influence over the investment decisions of investors. Also, comprehending investors' expectations and reactions to information flows in Bear and Bull period may help companies to make optimal investment plans for expansion (Ersan, 2008). Investment pattern of investors is impacted with the Bull and Bear market trends as these changing trends impact the investor sentiments which prompt them to enter/buy the market when the trend is Bullish and exit/sell when it is Bearish. Thus these trends should not be ignored by the investors and should be used for market timing by the investors. Studies like Guidolin & Timmermann (2005), Tu (2005) have pointed out the benefits of understanding and cost of ignoring the Bull and Bear phases which increases with the investment horizon.

Analysing the primary survey result shows that investor prefers more to invest in bull markets and the various factors which affect their decisions. In this section we try to empirically analyse the various strategies which can be used by investors depending on the current stock market phase.

The following objectives were further studied.

To investigate the effect of bull and bear markets on different investment strategy of the market timers and passive investors.

- I. It aims to examine whether alphas and/or betas in bull markets are not statistically different from alphas and/or betas in bear markets.
- II. To test whether the size based investment strategy yields extra normal returns in Bull and Bear phases.
- III. To test whether the value based investment strategy yields extra normal returns in Bull and Bear phase.
- IV. To test whether extra return is earned by formulating portfolios using Sharpe Index Single Model.
- V. To test whether the size premium and value premium is higher during bear or bull phases.

Stock portfolio and investment managers attract investors by claiming to earn positive alpha for their clients.⁷ However, a positive alpha is more desirable than a negative one by investors as investors prefer higher alpha. (Fabozzi & Francis, 1977) asserts that stock managers sell separate alpha and beta statistics for bull and bear markets for aggregate fees in commission from their clients. Fabozzi and Francis tested single-index market model for different time period distributed into bull and bear markets.

1. Investment Strategies based on Alpha and Beta in Bull and Bear Market (H₀₄)

The model used in this study is adopted from Black (1972) and Levy (1974) as used in Clinebell, Squires, & Stevens (1993) paper. To test whether investing in stock markets on the basis of Alpha and Beta is profitable to investors in Indian stock market, regression analysis is applied on stock returns. Regression analysis was done for model wherein returns on stock are regressed on market return (Sensex) and a dummy variable of bull/bear along with interaction between dummy variable and market Returns. In this study the significance and stability of alpha and beta over bull and bear markets is tested for the time period from January 2003 till September 2015 for the companies which were at any time during this period part of NSE Nifty 100 index. The results indicates that for 78.68% (69.85%) companies the alpha coefficient were not significant at 5% (10%) significance level. This

⁷ www.investopedia.com

implies that for most of the companies in NSE Nifty 100 index no abnormal returns were earned over the different phases of bull and bear and that alpha as a measure of performance remains stable over different market phases. Similarly the results for test of beta stability over bull and bear period show that for 92.64% (83.82%) stocks beta was not statistically significant over the bull and bear phase in Indian stock market.

2. Investing/trading during first six months or first three months during bull and bear phase respectively. (H₀₅)

Bry and Boschan Algorithm (1971) requires 6 months of subsequent returns to determine switching points, which implies turning points in stock markets cannot be identified immediately. This has important implications if the identification of turning points is to have value for investment decisions. The first 6 months of bull phase and first three months bear can therefore be tested to see if the returns in this period differ from the remaining months of each phase. For instance, if only the first 6 months of a bull market phase have significantly high returns, then it may be that by the time a turning point is identified, the majority of the gains to be had in a bull market have already occurred.

The strategy of investing/trading during first six months or first three months during Bull and Bear Market is checked through regression equation. The results indicate that the first 6 months of bull markets exhibit significantly higher returns than do the remaining months of bull markets, a higher significant return of 3.7% in first 6 months compared to 2.45% returns in remaining months. It is seen in previous chapter that total period return combined in all bull period is 6.18% monthly majority of which is realised in first six months of bull period, a finding that is consistent with Maheu, McCurdy, & Song (2012), Gonzalez, Powell, Shi, & Wilson (2005). Also, there is no significant difference in returns between the first three months and the remaining months of bear markets as the dummy coefficient is not statistically significant. While the intercept term is significant this shows that the return in remaining month of bear period is statistically significant. Thus in bear period the investor should hold the investment in stocks for longer duration so as to earn high return. The results suggest that the identification of turning points can potentially provide useful information to momentum investors who are interested in market timing, especially since bull markets usually last considerably longer than 6 months (the average is 21 months). All stages of a bear market are bad for investors; hence, more risk-averse investors might be interested in

avoiding bear markets once they are detected and follow long term investment, invest in T-bills and bonds, hold in fixed deposits and hold more cash.

3. Size and Value Investing during Bull and Bear Market. (H₀₆)

Timmermann and Quiros (2000) concluded that during recessions there are larger asymmetries in the risk and return attributes of smaller firms than those of larger firms. The behaviour of small and large firm's stock returns over different stock market phases namely bull and bear markets is analysed using the following dummy variable regression equation:

$$R_{pti} = \alpha + \beta_1(R_{mt} - R_{ft}) + \beta_2 D_t + \varepsilon_{it}$$

Where,

R_{p*ti*} = Return on portfolio , i= 1,2,3,4,5,

R_{mt} = Return on Market portfolio

R_{ft} = Return on Risk free rate, 91 days Government T-bills Rate

D_t = Dummy variable for Bull=1, Bear=0

α = intercept

β₁ = coefficient of market factor

β₂ = slope coefficient of dummy factor measuring the difference in risk adjusted returns during bull and bear market phases

ε_{it} = Random error term

To check the null hypothesis that there lies no significant difference between stock returns of size sorted portfolios during bull and bear periods, $\beta_2 = 0$, and the alternative hypothesis is that $\beta_2 > 0$ for smallest stocks portfolio. The regression analysis is applied on portfolios formed on the basis of Fama and French (Fama & French, 1996) (Fama & French, 2012) to test the hypothesis which is tested at 5% level of significance. The stock index constituents of CNX Nifty 100 index are adopted to formulate strategies on the basis of their market capitalization and book value to market value ratio. The monthly data is used for formulating portfolios on the basis of size and value factors for the study period from January 2003 till September 2015. The portfolio is formed by dividing the stocks into quintiles on the basis of stock characterization of size (market capitalization) and value (book value to market value). The market capitalization value and book value and market value of shares are taken as on financial year end each year, March end data.

Only those companies whose complete data for the study period was available were used. Those companies which were merged or suspended from trading were excluded. Thus every year data for different number of companies were available and used to formulate the portfolios.

The summary statistics of Average Size, Average BV/MV and Average Returns of Portfolios sorted on the basis of Size and BV/MV in Bull and Bear Phases in Indian stock market. The summary results show that all the size sorted portfolios have negative returns in bear period with variance decreasing from P1MC portfolio to P5MC portfolio. While in bull period all the size sorted portfolios are giving positive returns with variance decreasing from small (P1MC) portfolio to big (P5MC) portfolio. However, the SMB portfolio is negative in both bull and bear period but is less in bull period.

Similarly in value sorted portfolios earn negative returns in bear period and positive return in bull period. The value sorted portfolios, P1BVMV (4.0%) earn high return in bull period as compared to P5BVMV (2.6%) which gives a value premium of 1.4%. Thus the value stocks give high premium as compared to size sorted portfolios to investors in bull period.

The t-test statistics of monthly returns of portfolios in Bull and Bear phases- sorted on the basis of Size and BV/MV show significant values with significance level 5% for P1MC, P5MC and SMB portfolios for size sorted portfolio in bear period. Similar for bull period, size sorted portfolios P1MC, P3MC, P4MC, P5MC and SMB gives statistically significant returns with p-value 5%. The t-test statistics for value sorted portfolio show that all the portfolios P1BV/MV (High), P2BV/MV, P3BV/MV, P4BV/MV, P5BV/MV (low) have statistically significant mean returns in both the bull and bear periods. This implies value sorted portfolios give significant returns to investors in bull and bear period in Indian stock market.

Previous researchers have found that size is a factor which has effect on stock returns. The effect of size on stock return has positive effect and investors earn abnormal profit. Taking into account the effect of regime switching of Bull and bear market phases is tested on investment strategy on the basis of size. Thus, to analyse the economic viability of size effect if it gives supernormal returns in sub-periods, a strategy is adopted where investors should buy small stock portfolios (P1MC) and sell short large stock portfolio (P5MC). We assume that the investor or the trader will invest the proceeds of short selling in buying small stocks therefore his initial outlay cost will be Nil.

The t-values of size sorted portfolio for intercept, coefficient of market factor, i.e. beta and slope coefficient (β_2) of dummy factor which measures the difference in risk adjusted returns during bull and bear market phases is found to be statistically significant which shows that the returns on portfolio are statistically. This implies that the investors in Indian stock market earn statistically significant risk adjusted returns on their size sorted portfolios in bull and

bear phases. Thus incorporating the underlying current market phase in their portfolios gives superior returns to investors based on size sorted portfolio.

It can also be seen that as one moves from P1MC to P5MC the value of alpha (i.e. extra normal returns) declines sharply and for the large stocks portfolios it even turns negative (P3MC). This implies that investors earn statistically significant positive extra risk-adjusted returns on small stocks portfolios while return on the large stocks portfolios provide lower than what they should give their level of risk. These findings indicate the presence of strong size effect in bull and bear phase, i.e. small stocks outperform big stocks in different market phases.

The slope coefficient of the dummy variable is positive and statistically significant for the smallest stocks portfolios based on both size sorted and value sorted stocks. For P1MC the slope coefficient of the dummy variable is 0.048 (t-statistic 3.046) which indicates that P1MC portfolio generates a higher return during bear period than during bull period.

Similar for P1BVMV, the slope coefficient of dummy variable is 0.052 (t-statistics 2.93) is positive which shows that value sorted portfolios yields a positive rerun in value sorted portfolios in bear period.

To test whether the size and value premium is higher during bear or bull periods. (H₀₈)

In order to test whether size premium is sensitive to changes in underlying market conditions (bull or bear) we have used the following dummy variable regression –

$$SMB_t = \alpha + \beta * D_t + e_t$$

where

SMB_t = Size premium in month t

D_t = Dummy factor representing market conditions (bull or bear).

It takes value 1 during bull periods and 0 during bear periods.

α = Intercept term, it shows the size premium during bear periods

β = is coefficient of dummy variable which implies the excess of size premium earned during bull periods over bear periods.

In order to test whether value premium is sensitive to changes in underlying stock market conditions, i.e. bull or bear states we have used the following dummy variable regression –

$$HML_t = \alpha_2 + \beta_2 D_t + e_t$$

where

HML_t = Value premium in month t

D_t = Dummy factor representing market conditions bull and bear states. It takes value 1 in bull market while takes 0 in bear periods.

α_2 = is intercept term, shows the value premium during bear periods.

β_2 = is coefficient of dummy variable which implies the excess of value premium earned during bull periods over bear periods.

The empirical result shows that none of the slope coefficient of dummy factor is statistically significant for both size premium and value premium. Thus we accept the null hypothesis the size premium and value premium do not differ significantly over different market conditions bull and bear phase in Indian stock market during our study period. Thus small size or high value stocks are not providing any premium in bear period in Indian stock market.

4. Optimal Portfolio based on Sharpe's Single Index Model during Bull and Bear Market. (H₀₇)

The Sharpe model is based on the security's return relationship with the index return. The beta factor helps in determining the stocks systematic risk and unsystematic risk.

Thus, Sharpe Index Optimal portfolio model is an important model which selects the stocks on the basis of excess return to beta ratio.

$$\text{excess return} - \text{beta ratio} = \frac{R_i - R_f}{\beta_i}$$

Where, R_i = the expected return on stock i

R_f = the return on a risk-free asset

β_i = the expected change in the rate of return on stock i associated with one unit change in the market return

Sharpe single index model requires stock related input of stocks' beta, systematic risk, unsystematic risk, stock's return, risk free rate of return such as on government security or Treasury bill, expected return and variance on market index. The selection of stocks used to create Sharpe single index portfolio is done from CNX Nifty 100 index constituents. The index prices of BSE Sensex, comprising of top 30 stocks by their market capitalization, was taken as proxy for market return and 91 days Government T-Bills rate was adopted as risk-free rate to create portfolios and risk-return relation and beta for stocks. The stock price data and market index data is taken from the period January 2003 till September 2015.

The Sharpe's single index model is vital for stock selection during bull and bear period as it selects stock whose excess return to beta ratio is high. The excess return to beta ratio shows return of stock per unit of its the systematic risk or non-diversifiable risk. This helps the investor to select those stocks which will give positive return to investors even during Bear period. The summary of stocks selected to form portfolio in each Bear phase shows that the investors would earn positive return in each bear phase except for Bear phase IV. The portfolio formed on basis of stocks selection in bear phases gives positive and higher return than the market index which correspondingly gives negative returns in these phases. It can be

seen that only in Bear phase from January 2004 till May 2004 gives us a negative return. The portfolio beta is also less than 1 which means that the portfolio will fluctuate less than the variability in the market prices.

Also in the Bull phase the Sharpe's single index model is applied in similar method. The results showed the number of stocks in each portfolio formed in each bull phase, market variance and return is compared with the return and variance of the portfolio formed. The portfolio formed in the first bull phase which is from May 2003 till December 2003 contains 25 stocks which gives portfolio return of 13.48% with total risk of 10.3%. This return is higher than that of market return of 8.95%. The beta of portfolio of bull phase I is 0.54 which means that the portfolio return will change in less proportion to the market. However, in the table it can be seen that portfolio return is higher than the market return. Thus formulating portfolio in bull phase on the basis of Sharpe's single index model gives higher return to investor.

Conclusion & Implications

The underlying current market phase has important implications for investors which has been analysed in the study. Not many studies in Indian researches have attempted to make an attempt to study the effect of market phase bull and bear on investment decisions of Indian investors. The study identifies seven bear phases and six bull phases in Indian stock market during the time period April 1995 till March 2016. In times of recession and economic crisis the sentiment of investors become negative leading to downfall in stock market. This downward trend is often termed Bearish by traders, wherein the investors lose faith in the market and large number of selling takes place. The relation between market phases preferred by individual and different behavioural factors show that market phase has a statistically significant relation with trading attitude, disposition effect, confidence, risk tolerance of an individual. Thus, this confirms that understanding and incorporating bull and bear phases in the investment strategies of an investor have economic implications and that trading attitude of an individual changes with the underlying market condition. In bull period the market sentiments are positive and economy is growing with increased production. Thus the prices of share keep increasing and reach new high levels. Thus in bull period trading attitude is positive with more individuals trading or entering the stock markets. However, in bear period the market sentiments are negative and economy is not growing with decreased production and consumption. The stock market decreases due to negative sentiments which create a

sense of fear in the investor which keeps the investor out of the market. Disposition effect, confidence, risk tolerance of an individual also influences the investment decisions depending on the underlying market phase. Risk tolerance of an investor increases in bearish period but decreases during bull period. Disposition effect has been seen to have higher effect during bearish period in Japanese investors (Cheng, Lee, & Lin, 2013). The differences in market volatility which is higher during bear markets than in bull markets influences the disposition effect in investors. Important behavioural studies by Kahneman and Tversky (1979), Thaler (1985) explores the psychological biases which provide possible explanations for investor behaviour and for outstanding asset pricing anomalies such as the equity premium puzzle, the value premium, and the momentum effect. The study adds new dimensions to stock market behaviour using primary survey and analysing investment strategy for Indian investors. The behavioural factors which affect the investment decisions of Indian investors were found to be trading behaviour, self-attribution, trading attitude, disposition effect, over-confidence, risk tolerance, herd behaviour.

The results of primary responses creates a paradigm wherein the investment is not solely seen as a function to generate returns rather respondents found it as activity in different time sets and domains, including insurance and tax saving. The empirical findings makes it evidentiary that the investments undertaken during the bear run or even on the onset of bull run, do gives higher returns. Despite the availability of higher returns the investors' are less driven for the systematic and regular investment in the markets. The researchers would strongly recommend that the systematic investment plans and investment in all mutual funds should provide tax benefit to the investor catering the larger behaviour bias.

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